Nothing:
The First Object

Diegο Rincón
Independent

Abstract

Nothing is not the absence of objects. It is the first object. $\emptyset$, the empty set, requires no construction: no prior set, no elements, no grammar [2] beyond the axiom that says it exists. From $\emptyset$ alone, all of mathematics is built: every number, every function, every space is a structured arrangement of nothings. Zero is the numerical $\emptyset$: not no quantity but the quantity that is the ground of all quantity. The unit () is the first step from nothing: the minimal something, the type with one value, the object that distinguishes presence from absence. The Wall [1] between nothing and something is the first Wall — the crossing that cannot be explained, only noted. Everything is on one side. $\emptyset$ is on the other. The universe began when something crossed it.

1 The Empty Set

Definition 1 (Nothing as object)**.** The empty set $\emptyset$ is the unique set with no elements:

$$\forall x,\ x \notin \emptyset.$$

It is not the absence of a set. It is a set — the set that contains nothing. This distinction is everything: $\emptyset$ is an object, not a void. It has properties, participates in relations, is the initial object in the category of sets: there is exactly one function $f : \emptyset \to X$ for every set X — the empty function, vacuously defined, requiring no cases.

$\emptyset$ requires no construction. It is the axiom of the void: one existence claim, no content. Every other set requires more: elements, operations, structure built on prior structure. $\emptyset$ is prior to all of it. It is the proper design [3] for the empty domain: the one object that was there before anything else was there.

Theorem 2 (All mathematics from nothing)**.** *Von Neumann's construction builds all mathematics from $\emptyset$:*

$$\begin{aligned} 0 &= \emptyset \\ 1 &= \{\emptyset\} \\ 2 &= \{\emptyset,\ \{\emptyset\}\} \\ 3 &= \{\emptyset,\ \{\emptyset\},\ \{\emptyset, \{\emptyset\}\}\} \\ n+1 &= n \cup \{n\}. \end{aligned}$$

Every natural number is a set of nothings arranged in a structure. Every integer is a pair of natural numbers. Every rational is a pair of integers. Every real is a Dedekind cut of rationals. Every complex number is a pair of reals. Every function is a set of ordered pairs. Every ordered pair $(a, b) = \{\{a\}, \{a, b\}\}$ *is a set.*

Everything is, at bottom, a structured arrangement of $\emptyset$*. The universe of mathematics is* $\emptyset$ *reflected in itself, nested in itself, arranged and re-arranged in every possible structure. One object. All of mathematics.* $\emptyset$ *is the proper design [3] for all of it.*

2 Zero

Proposition 3 (Zero as numerical nothing)**.** *Zero is the numerical* $\emptyset$*: the additive identity, the quantity that adds nothing, the number that is the ground of all numbers.* $0 + n = n$*. Zero does not change what it touches. It is already there before any quantity arrives.*

Zero was the last number discovered, not the first. Ancient mathematicians counted things: sheep, days, soldiers. Zero requires a different insight — not the count of things present but the count of things absent. The count of absent things is not nothing. It is zero: a precise, definite number, the number that marks the place where counting has not yet begun.

The discovery of zero was the discovery that nothing is an object. Not a placeholder, not an absence, not a silence — a number, the first number, the number from which all counting departs.

3 The Unit

Proposition 4 (One step from nothing)**.** *The unit* $() = \{*\}$ *is one step from* $\emptyset$*: the minimal something, the set with exactly one element. In type theory,* $()$ *is the unit type: the type with exactly one value (itself), the type that carries no information beyond existence.*

$\emptyset$ *and* $\{*\}$ *are dual:*

- $\emptyset$ *is the initial object in* **Set***: one function out, to every set.*
- $\{*\}$ *is the terminal object in* **Set***: one function in, from every set.*

From nothing ($\emptyset$*), everything can go. To the unit (*$\{*\}$*), everything can come. Nothing and unit are the two poles of the category: the origin and the destination, the void and the minimum, the empty and the singular.*

The integers are the distance between them, extended in both directions by the same step: $0 \to 1 \to 2 \to \cdots$ *and* $0 \to -1 \to -2 \to \cdots$ *The line of numbers is the expansion of the gap between* $\emptyset$ *and* $\{*\}$*, built from the single step that crosses it.*

4 The First Wall

Proposition 5 (Nothing to something: the uncrossable crossing)**.** *The Wall [1] between* $\emptyset$ *and* $\{*\}$ *is the first Wall — more fundamental than the Arithmetic Wall, more fundamental than the Turing Wall, more fundamental than the Wall of consciousness.*

Every other Wall is a Wall between two somethings: two grammars, two languages, two objects that cannot reach each other. The first Wall is a Wall between nothing and something: the crossing that cannot be derived from prior structure because there is no prior structure.

No grammar [2] generates this crossing. No design [3] precedes it. No a priori language [4] conditions it. The step from $\emptyset$ to $\{*\}$ is the one step that is not a consequence of anything. It is the axiom of existence: something rather than nothing, not because something is better but because something is.

The universe began when this Wall was crossed. Every subsequent Wall is a face of that first crossing, viewed from inside the something that resulted.

5 The Zeros as Nothing

Remark 6 (Where $\zeta(s) = 0$)*.* The non-trivial zeros of $\zeta(s)$ are the nothing-places of the Riemann zeta function: the points where the function vanishes, where the arithmetic grammar [1] falls silent, where the spectral string has no energy.

The zeros are not voids. They are objects — precise, located, countable. Like $\emptyset$, they are a structured nothing: not the absence of a zero but the zero, the point where the function is exactly $\emptyset$.

The Riemann Hypothesis is the claim that all the nothing-places are on one line. The zeros are arranged the way $\emptyset$ is arranged in the Von Neumann construction: not scattered, not random, but structured by the same grammar that built them from nothing.

The proof of RH [7] is the proof that the zeros' nothingness has a particular shape — that where the function is $\emptyset$, it is $\emptyset$ for a reason, and that reason is the critical line.

The nothing is not arbitrary. It is designed [3]. The zeros are the most structured nothing in mathematics.

Remark 7 (The ground state)*.* The ground state of a quantum system is the state of minimum energy — not zero energy (that would violate the uncertainty principle) but the irreducible minimum. The ground state is the physical $\emptyset$: not nothing, but the least something possible.

The aura [5] at ground state: all bonds closed, no open valences, the completing field fully present, no crisis, no excess, no further reduction possible. The organism at the ground state of its aura is not empty — it is the minimal full.

The still lever [6] is the ground state of dynamics: zero net force, zero work, zero motion. Not the absence of the lever but the lever at rest.

Nothing, in every domain, is the ground state of that domain: the irreducible minimum, the object before all objects, the first design, already present, requiring nothing.

References

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[5] Rincón, D. (2026). Five Walls, One Aura. Preprint.

[6] Rincón, D. (2026). Mathematics. Preprint.

[7] Rincón, D. (2026). Week 10: Self-Adjointness. Preprint.

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